

Aim

To design and demonstrate a **pyramidal AI approach** (neural networks + generative AI + optimization) that forecasts rooftop solar generation and household demand and uses those forecasts as inputs to an optimization/dispatch layer that adaptively schedules residential battery charging/discharging and household loads — enabling robust, real-time, utility-/household-level management of PV + battery energy.

Objectives

1. Build ML models (ANN regression, LSTM) to forecast solar generation and household electricity demand.
2. Develop a conditional Generative Adversarial Network (solarGAN) that jointly simulates weather, PV output and demand time series suitable for short-term forecasts and planning.
3. Combine forecasts with an optimization layer to produce optimal battery dispatch schedules (24-hour ahead) for peak shaving.
4. Propose a real-time, online-learning deployment that continuously adapts model parameters and decisions as conditions change.
5. Demonstrate improved system performance (reduced charge/discharge deviations, cost savings, improved reliability) from the AI approach.

Introduction

- Rapid adoption of rooftop PV + residential batteries creates many distributed energy resources (DERs). Managing them safely and profitably requires intelligent coordination because generation is intermittent and demand patterns vary. Smart, dynamic control at household scale can: reduce customer bills, provide grid services (peak-shaving), and increase renewable utilization. However, current solutions are often static (averaged policies, blanket export limits) and fail to adapt to local, rapidly changing conditions. The paper argues advanced ML (ANN, LSTM, GANs) and optimization can create **real-time adaptive** controllers to safely scale DER deployment.

Methodology

1. **Data preprocessing & feature engineering**
 - Collect smart meter-like measurements: PV generation, household active power (main breaker monitoring), ambient/weather features. Fit AC-side generation using fitting functions; derive net export/import by subtracting fitted net loads.
2. **Forecasting models**
 - **ANN regression** for short-term regression tasks (forecasting, classification of good/bad operating states).

- **LSTM (Long Short-Term Memory)** for time-series forecasting of solar generation and load demand (captures temporal dependencies).
- **Conditional GAN (solarGAN)** jointly generates/predicts weather + PV + demand time-series at required resolution to support planning and sizing tasks. The GAN is designed to be trainable from limited real-world series.

3. Signal transforms & hybrid methods

- Wavelet transforms used to decompose time series prior to ML (improve modeling of nonstationary signals).
- Quantum-inspired genetic algorithms and region elimination techniques used for optimization/optimal sizing.

4. Optimization & dispatch

- Forecasts feed into an optimization problem (a cost function linear in PV and storage contributions in the simple form they used). The optimizer provides 24-hour ahead dispatch schedules and decisions for peak shaving. They implement a “simplified neural optimizer” (a neural-network-based solver for constrained optimization problems).

5. Online learning & deployment

- The models support online update / online learning to adapt to changing conditions.

System architecture

Main components (in order of data flow):

1. Data Layer

- Inputs: PV inverter telemetry, household main breaker (load) data, weather station / forecast feeds, contextual metadata (time of day, tariff/TOU).
- Preprocessing: cleaning, wavelet transform, feature extraction, normalization.

2. Forecast Layer

- LSTM and ANN models produce short-term solar and demand forecasts.
- solarGAN generates synthetic/conditional series for planning and robust estimation.

3. Predictive Analytics & Health

- Anomaly detection and predictive maintenance inputs — monitors PV and battery health. (Paper suggests PV diagnostics and recommender functions.)

4. Optimization / Decision Layer

- Neural optimizer + classical solvers compute battery charge/discharge schedules, device-scheduling recommendations, and buy/sell actions (time-of-use aware). Outputs include prioritized load scheduling and battery setpoints.

5. Execution Layer / Edge Control

- Sends actuator commands to inverters, battery management system (BMS), device control (if home automation integrated). Also pushes recommendations/alerts to user dashboard.

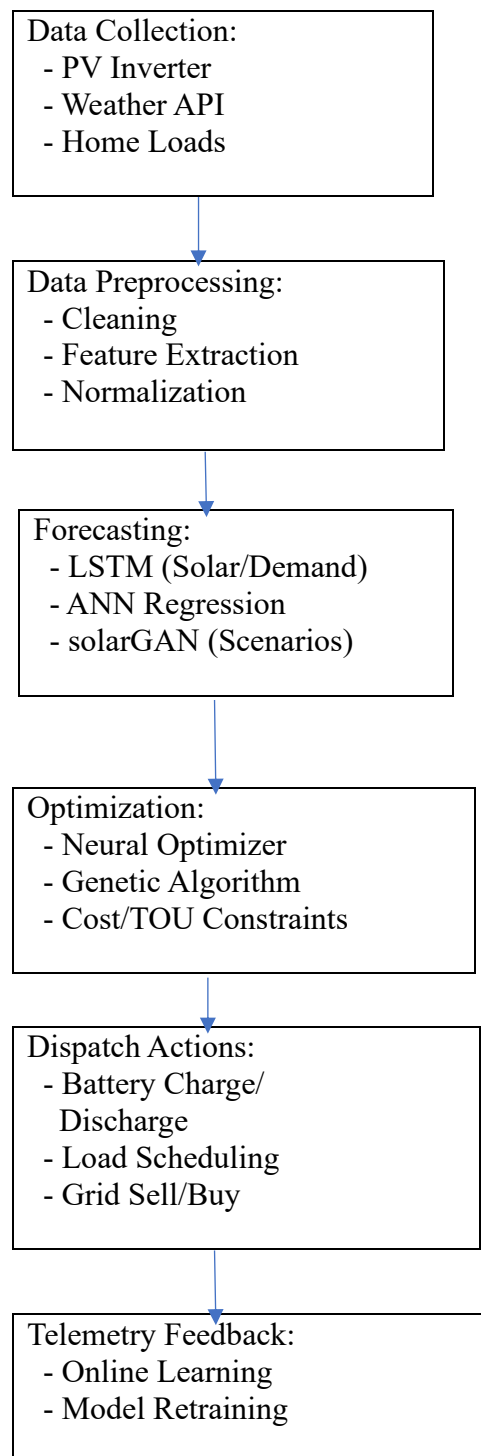
6. Online Learning Feedback

- Actual telemetry returns to data layer to retrain/update models continuously (closing the loop).

Algorithms used — list & roles

- **Artificial Neural Networks (ANN)** — regression, classification (performance estimation, quick decision proxies).
- **LSTM (Recurrent Neural Network)** — time-series forecasting of solar generation and household demand.
- **Conditional Generative Adversarial Network (solarGAN)** — generate joint weather/PV/demand scenarios and augment training data.
- **Wavelet transforms** — time-frequency decomposition for nonstationary signal preprocessing.
- **Quantum-inspired Genetic Algorithm / region-elimination techniques** — optimization for battery sizing and schedule planning.
- **Neural optimizer (NN-based solver)** — lightweight supervised NN that approximates or solves constrained optimization problems fast.
- (Mentioned areas) **Deep reinforcement learning** — suggested for battery dispatch in time-varying environments (paper cites potential suitability).

Flowchart / Block Diagram



Claim / Results — key findings (as reported)

- The proposed **real-time multi-task model** (with online learning and optimization) delivers **dynamic improvements** in operation, showing **reduced charge/discharge deviations** and better cost/performance in tests. The paper states the AI model “has shown an average of real-time dynamic improvement from online learning across all of the tests conducted.”

- The solarGAN approach can **jointly forecast weather, rooftop PV and household demand** at time resolutions useful for network planning, identifying households with PV, and scheduling controllable loads.
- The simplified neural optimizer demonstrated **fast, effective** solutions for a class of linearized cost functions combining PV and storage, with strong performance in example scenarios.

Limitations

1. **Quantitative details are limited** — the paper reports “improvements” and “average real-time dynamic improvement” but does not present a comprehensive set of numerical benchmarks (e.g., % energy saved, reduction in cost, sample sizes, statistical confidence intervals) in the sections provided. That makes rigorous performance comparison hard.
2. **Dataset description & reproducibility** — there is not enough detail about dataset size, sampling rates, geographic diversity, or public availability of the data used to train/test solarGAN / LSTM / ANN. This hinders reproducibility.
3. **Scalability & communication constraints** — while they recommend cloud-based recommendation engines, real-world rollouts face latency, privacy, smart-meter coverage and telecom limitations (the paper notes incomplete smart meter rollout).
4. **Economic modeling & tariff variability** — the optimization treats cost functions in simplified linear forms. Time-varying tariffs, dynamic markets, and transaction costs could make optimization more complex in practice.
5. **Safety & grid interaction details** — integration into distribution network protection schemes, safety interlocks and grid operator constraints are mentioned conceptually but not evaluated in hardware-in-the-loop or field trials.
6. **Generality across battery chemistries and BMS** — control strategies tuned for lithium-ion may not directly map to sodium-ion or other chemistries; further experimentation needed.